



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / First Semester – Spring 2022

Statistics-I

Course Code: STAT-101

Roll No.

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (15x2=30)

- Differentiate between Parameter and Statistic
- Define qualitative variable and quantitative variable, give any one example of each type.
- Describe briefly the main steps in the preparation of Frequency table from raw data.
- What is a cumulative frequency curve? How does it differ from the ordinary curve?
- Show that $\sum (x_i - a)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - a)^2$ where 'a' is any arbitrary number.
- Differentiate between simple Arithmetic Mean and weighted Arithmetic Mean.
- What is meant by skewness? What are the measures of skewness, name them and also give formulae.
- The following values have been obtained from a frequency distribution of weights (lb) having 125 observation after making the substitution. $x = 16 + 5U$, $\sum fu = -46$, $\sum fu^2 = 306$, $\sum fu^3 = -242$, and $\sum fu^4 = 1962$ check that either the distribution is Meso Kurtic or not.
- Compare the simple Index and composite Index.
- Show that Fisher's ideal Index satisfies the time reversal test.
- Show that coefficient of determination r^2 is $r^2 = \frac{a\sum y + b\sum xy - (\sum y)^2/n}{\sum y^2 - (\sum y)^2/n}$
- Describe the properties of regression line.
- What is analysis of time series.
- Write brief note on semi average method.
- Define standard deviation of regression.

Solve the following questions.

Q:2 (a) Find (i) Arithmetic Mean and (ii) Geometric Mean of the series. 1, 3, 9, 27, 81, 3^n

(b) The mean and Variance of variable "x" are 60 and 64 respectively. Find Mean and Variance of new variable if

(i) All the Values of x are increased by 20 points.

(ii) All the values of x are increased by 10%.

(5+5=10)

Q:3 (a) Obtain (i) Simple aggregative value index. (ii) An index of average value, for each year from the following data.

Years	Declared values	Values on the basis of 2010 values
2010	900	900
2011	1050	988
2012	1500	1000
2013	1700	832

(b) For n pairs of value of two variables when each variable is ranked in order (1 to n) show that

$$\text{coefficient of rank correlation is } r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \quad (5+5=10)$$

Q:4 (a) Compute Seasonal Indices by link Relative Method for the following data.

Year	Jan-March	April-June	July-Sep	Oct-Dec.
2001	80	101	126	92
2002	85	110	123	90
2003	83	99	125	94
2004	86	102	129	92

(b) Fit a Straight line $y = a + bx$ from the following results for the year 1948-58 (Both inclusive)

$$\sum x = 0, \sum y = 438.9, \sum x^2 = 110$$

$$\sum xy = -84.4$$

(6+4=10)

Find out the estimated trend values for 1960 and 1961.

Q:3 (a) Calculate the coefficient of correlation from the following data:

X	3	4	5	6	7	8	9	10	11	12
Y	25	24	20	20	19	17	16	13	10	6

(4+6=10)

(b) Construct index numbers from the following data by applying (i) Laspeyre's method (ii) Paasche's method.

Commodities	Base Year		Current Year	
	Prices	Quantities	Prices	Quantities
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

Q:4 (a) Compute Seasonal Indices by link-relative method from the following data.

(6+4=10)

Year	Quarters			
	I	II	III	IV
2001	112	125	122	110
2002	119	132	147	115
2003	120	142	150	118
2004	128	151	162	125

(b) Fit a Straight line $y = a + bx$ from the following results

$$\sum x = 0, \sum y = 438.9, \sum x^2 = 110$$

$$\sum xy = -84.4$$

for the year 1948-58 (Both inclusive). Find out the estimated trend values for 1960 and 1961.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- (i) Define population and sample.
- (ii) Define tabulation. What are the necessary parts of a table?
- (iii) For which type of data, the Geometric mean and harmonic mean are suitable?
- (iv) The mean of 10 numbers is 8. If an eleventh number is included, the mean becomes 9. What is the value of the eleventh number?
- (v) Define the term dispersion. Write any three absolute measures of dispersion.
- (vi) Differentiate between symmetry and skewness.
- (vii) If sum of squared deviation from mean for 80 values is 1600, what will be the variance standard deviation and coefficient of variation if $\bar{x}=20$
- (viii) Differentiate between simple and composite index numbers.
- (ix) Differentiate between un-weighted index and weighted index number.
- (x) Differentiate between the term regression and correlation.
- (xi) What is the relationship between regression coefficients and correlation coefficients? Find the correlation coefficient, if the regression coefficients are 1.2 and 0.60.
- (xii) What is meant by coefficient of determination?
- (xiii) Define the term rank correlation; also write down the formula of rank correlation.
- (xiv) Define the term time series. Write down the components of a time series.
- (xv) Differentiate between cyclical variations and sessional variations.

Solve the following questions.

Q:2 (a) The following data relate to size of shoes sold at a store during a given week. Find the median and the modal size of shoes also find Q_1 and P_{90}

(6+4=10)

Size of Shoes	5	$5\frac{1}{2}$	6	$6\frac{1}{2}$	7	$7\frac{1}{2}$	8	$8\frac{1}{2}$	9	$9\frac{1}{2}$
No of Pairs	3	7	20	30	60	50	25	17	8	2

(b) An analysis of the monthly wages paid to workers in two firms A and B belonging to the same industry gives the following results:

	Firm A	Firm B
No. of wage earners	120	145
Average monthly wages (Rs.)	28620	29540
Variance of wages	1250	1400

- (i) Which firm A or B pays the larger amount of monthly wages?
- (ii) In which firm A or B is there greater relative variability in individual wages?



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : First Semester – Fall 2021

Paper: Statistics-I

Course Code: STAT-101

Roll No.

Time: 3 Hrs. Marks: 60

Q.1. Answer the following short questions. (15x2=30)

- Define Descriptive and Inferential statistics
- Define a variable. Differentiate between discrete and a continuous variable.
- What is meant by frequency distribution and cumulative frequency distribution.
- Define the statistical term histogram and frequency polygon.
- What is a measure of central tendency? What are its desirable qualities?
- Show that $\sum(x_i - \bar{x}) = 0$
- What is coefficient of variation? What purpose does it serve?
- The first four moments of a distribution about $x=2$ are 1, 2.5, 5.5 and 16 calculate the first four moments about mean and calculate β_1 and β_2 .
- Compare the simple aggregative and weighted aggregative index numbers.
- Prove that fisher's ideal index satisfies the factor reversal test.
- Show that standard deviation of regression $sy.x = \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n-2}}$
- Define rank correlation, which quantity is added to $\sum d^2$ for tied ranks.
- What is meant by coefficient of determination?
- Distinguish between the secular and seasonal variation in time series analysis.
- Define detrending and deseasonalization.

Solve the following questions.

Q:2 (a) The following data relate to size of shoes sold at a store during a given week. Find the median and the modal size of shoes also find D_9 and P_{70} .

(6+4=10)

Size of Shoes	5	$5\frac{1}{2}$	6	$6\frac{1}{2}$	7	$7\frac{1}{2}$	8	$8\frac{1}{2}$	9	$9\frac{1}{2}$
No of Pairs	3	7	20	30	60	50	25	17	8	2

(b) Show that $\text{var}(x+a) = \text{var}(x)$ where 'a' is some constant.

Q:3 (a) Show that correlation coefficient is independent of change of origin and scale i.e. $r_{xy} = r_{uv}$ if

$$u = \frac{x-a}{h}, v = \frac{y-b}{k}$$

(4+6=10)

(b) Construct fisher's ideal index number for 2011, taking 2010 as base

Items	Quantity		Value	
	2010	2011	2010	2011
A	110	175	600	1000
B	90	120	400	800
C	70	85	200	350

Q:4 (a) Compute the seasonal indices by ratio to trend method using least square line $y=119.56+2.76x$ by taking origin in the centre and unit of x being half quarter.

(b) Deseasonalize the values of years 1999 and 2000

(7+3=10)

Year	QUARTERS			
	I	II	III	IV
1997	72	98	79	106
1998	79	122	101	143
1999	94	141	128	160
2000	125	143	135	187

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

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- (i) Define population and sample.
- (ii) Define tabulation. What are the necessary parts of a table?
- (iii) For which type of data, the Geometric mean and harmonic mean are suitable?
- (iv) The mean of 10 numbers is 8. If an eleventh number is included, the mean becomes 9. What is the value of the eleventh number?
- (v) Define the term dispersion. Write any three absolute measures of dispersion.
- (vi) Differentiate between symmetry and skewness.
- (vii) If sum of squared deviation from mean for 80 values is 1600, what will be the variance standard deviation and coefficient of variation if $\bar{x} = 20$?
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- (xii) What is meant by coefficient of determination?
- (xiii) Define the term rank correlation; also write down the formula of rank correlation.
- (xiv) Define the term time series. Write down the components of a time series.
- (xv) Differentiate between cyclical variations and sessional variations.

Solve the following questions.

Q.2 (a) The following data relate to size of shoes sold at a store during a given week. Find the median and the modal size of shoes also find Q_1 and P_{90} . (6+4=10)

Size of Shoes	5	$5\frac{1}{2}$	6	$6\frac{1}{2}$	7	$7\frac{1}{2}$	8	$8\frac{1}{2}$	9	$9\frac{1}{2}$
No of Pairs	3	7	20	30	60	60	25	17	8	2

(b) An analysis of the monthly wages paid to workers in two firms A and B belonging to the same industry gives the following results:

	Firm A	Firm B
No. of wage earners	120	145
Average monthly wages (Rs.)	28620	29540
Variance of wages	1250	1400

- (i) Which firm A or B pays the larger amount of monthly wages?
- (ii) In which firm A or B is there greater relative variability in individual wages?

Q:3 (a) Calculate the coefficient of correlation from the following data.

X	3	4	5	6	7	8	9	10	11	12
Y	25	24	20	20	19	17	16	13	10	6

(4+6=10)

Q:4 (b) Construct index numbers from the following data by applying (i) Laspeyre's method (ii) Paasche's method.

Commodities	Base Year		Current Year	
	P_0 Prices	Q_0 Quantities	P_1 Prices	Q_1 Quantities
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

Q:4 (d) Compute Seasonal Indices by link-relative method from the following data.

(6+4=10)

Year	Quarters			
	I	II	III	IV
2001	112	125	129	110
2002	119	132	147	115
2003	120	142	150	118
2004	128	151	162	125

(b) Fit a Straight line $y = a + b \cdot x$ from the following results

$$\sum x = 0, \sum y = 438.9, \sum x^2 = 110$$

$$\sum xy = -84.4$$

for the year 1948-58 (Both inclusive). Find out the estimated trend values for 1960 and 1961.



UNIVERSITY OF THE PUNJAB

First Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Statistics-I

Course Code: STAT-101 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q2. Explain the following

(4x5)

- Measures of Central Tendency
- Measures of Dispersion
- Index Numbers
- Properties of Correlation Coefficient
- Components of Time Series

Q3. (a) A distribution consists of 3 components with frequencies 45, 40 and 65, (03)
 having their means 2, 2.5 and 3 respectively. The standard deviations of
 the three components are 1.5, 2.0 and 2.5 respectively. Find the combined
 standard deviation.

(b) The following data gives the runs made by two players in their 10 innings. (06)

Player A	40	45	60	10	75	20	60	35	42	100
Player B	45	42	38	40	35	41	50	42	39	48

- Which player is better on average run getter?
- Which player is more consistent than the other player?

Q4. The following table represents the death rate (y) in road accidents (per 100) (06)
 and the maximum speed limit (x) miles per hour) in different
 countries.

Y	3.0	3.3	3.4	3.5	3.6	4.3	4.7	4.9	5.1	6.1
X	55	55	55	70	55	60	55	60	60	75

- Estimate a linear regression on model Y on X from the data.
- What would be the expected death rate when speed limit is 65 miles per hour?

P.T.O.



UNIVERSITY OF THE PUNJAB

First Semester 2018
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistics-I
Course Code: STAT-101

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SHORT QUESTIONS

Q2. Explain the following:

(4 marks each)

- Properties of Arithmetic Mean
- Properties of Variance
- Properties of Correlation Coefficient
- Index Numbers
- Components of Time Series.

SUBJECTIVE

Q3 (a) From a random sample of voters in Rawalpindi, Islamabad, voters are classified by age group, as shown by the following data. (06)

Age Group	18-24	24-30	30-36	36-42	42-48	48-54	54-60	60-66	66-72
Frequency	170	220	260	350	330	300	320	210	150

Calculate Mean and Geometric Mean of the above data.

(b) Following table gives the birth rates and death rates per hundred thousand of a few districts in Punjab. Represent them in a Component bar chart. (04)

Country	Birth Rate	Death Rate
Lahore	330	240
Multan	300	190
Faisalabad	180	80
Gujrat	250	160

Q4 Compute first four Mean moments of the following Data. Also compute coefficient of skewness and kurtosis. (10)

Weight (grams)	65-84	85-104	105-124	125-144	145-164	165-184	185-204
Frequency	90	100	170	100	50	40	50

Q5. (a) The following are the results of height (X) and weight (Y) of 12 persons. (05)

$$\sum x = 766, \quad \sum y = 1700, \quad \sum xy = 109380, \quad \sum x^2 = 49068, \quad \sum y^2 = 246100$$

Estimate a regression line of Y on X from the above data. Also find the correlation coefficient between height and weight.

(b) Calculate the seasonal indices by ratio to moving average method from the following data. Also deseasonalize the data for the year 1944. (05)

Year	Jan	Feb	Mar	April	May	June	July	Aug	Sep	Oct	Nov	Dec
1943	6.2	1.8	0.9	1.4	3.2	2.3	2.2	3.2	3.4	2.7	2.1	2.1
1944	3.3	1.7	0.5	2.2	1.5	2.5	2.8	3.2	4.2	4.5	6.1	2.8